Ashab Uddin

uddin81@uwindsor.ca | ashab.uddin76@gmail.com | +1-226-975-6371 | Portfolio | LinkedIn | Google Scholar

Professional Summary

Electrical and Automation Engineer and PhD Candidate in Electrical Engineering with 10+ years of industrial experience in plant automation, process control, electrical maintenance, troubleshooting, commissioning, and optimization of high-speed manufacturing systems, complemented by 4+ years of research in AI, intelligent control, reinforcement learning, and distributed optimization. Hands-on expertise in PLCs, SCADA, DCS, VFDs, servo drives, industrial instrumentation, and communication networks, with experience applying Python and PyTorch to intelligent control and cyber-physical systems. Bridges extensive industrial engineering experience with advanced AI and optimization methods for next-generation automation, intelligent manufacturing, energy-efficient operation, and complex cyber-physical systems.

Technical Skills

PLC & Control Platforms: Siemens S7-300/400/1200/1500, Allen-Bradley ControlLogix, B&R X20; STEP 7, TIA Portal, SIMATIC Manager, Studio 5000

SCADA & HMI: WinCC RT Professional, FactoryTalk View, Zenon, Siemens KTP HMI

Drives & Motion Control: VFDs, DC Drives, Servo Drives; ABB ACS355/ACS550/ACS880 and DCS880, Siemens SINAMICS, Danfoss FC-series, Yaskawa V1000, B&R ACOPOS

Industrial Communication: PROFINET, PROFIBUS, Industrial Ethernet, Modbus TCP/IP, CAN, SERCOS III, EtherCAT, AS-i, PLC Networking

Maintenance & Reliability: Preventive, Corrective and Breakdown Maintenance, Electrical Troubleshooting, PLC Fault Diagnostics, Root Cause Analysis (RCA), Equipment Reliability, Maintenance Planning

Instrumentation & Process Control: RTD, Thermocouple, Load Cell, Flow, Level and Pressure Transmitters, Control Valves, PID Control, Parameterization and Loop Tuning, Temperature and Pressure Control, Process Monitoring

Plant & Electrical Engineering: Commissioning, Project Management, Electrical Installation, Substation Systems, SLD Review, Fire Protection Systems, Audit & Inspection, Electrical Safety

Maintenance Systems: Oracle EBS Enterprise Asset Management (EAM), Maintenance Documentation

Engineering Tools: Python, MATLAB, Simulink, C++, AutoCAD, Git, Linux

Current Research Experience

Graduate Researcher – AI, Intelligent Control & Optimization 2022 – Present

University of Windsor *Windsor, Canada*

- Develop AI-based decision-making and intelligent-control frameworks using Python and PyTorch for dynamic cyber-physical and real-time systems.
- Research deep reinforcement learning, multi-agent systems, hierarchical control, and distributed optimization for resource allocation, energy efficiency, and constraint-aware system-performance optimization.
- Design simulation environments and evaluation frameworks to analyze system behavior, operational trade-offs, and control performance under dynamic conditions and limited resources.
- Develop priority-aware, multi-agent, and hierarchical learning architectures that decompose complex engineering problems into coordinated strategic and decentralized control decisions.
- Published first-author research in IEEE Transactions and international IEEE conferences in reinforcement learning, intelligent control, optimization, and distributed intelligent systems.

Industrial Experience

Assistant Manager – Electrical & Automation 2018 – 2022

Transcom Beverages Ltd. (PepsiCo Franchisee)

- Led electrical maintenance, automation, process-control, and troubleshooting activities for high-speed beverage production systems, including 600 BPM KRONES CSD, 400 BPM KHS/TFG CSD, hot-fill lines, pasteurizers, pallet-wrapping systems, CO_2 processing, and HVAC equipment.
- Led and supervised a multidisciplinary maintenance team of up to 35 engineers and technicians, managing manpower allocation, preventive and corrective maintenance, shift coverage, breakdown priorities, spare-parts coordination, equipment improvements, and commissioning support.
- Coordinated maintenance and engineering activities with production, quality, safety, vendors, contractors, and OEM representatives, including SIPA, SMI, KHS and KRONES technical teams, to resolve technical issues, minimize downtime, and maintain production continuity.
- Performed PLC, VFD, servo-drive, DC-drive, and industrial-network fault diagnostics across Siemens S7-300/400/1200/1500, Allen-Bradley 1756 ControlLogix, B&R X20, ABB ACS-series, ABB DCS880, Siemens SINAMICS, Danfoss FC-series, Yaskawa V1000, and B&R ACOPOS platforms.
- Diagnosed, modified, and optimized automation and process-control systems using TIA Portal, WinCC Professional, Studio 5000, FactoryTalk View, and Zenon, and troubleshot PROFIBUS, PROFINET, Industrial Ethernet, CAN, SERCOS III, EtherCAT, and AS-i networks.
- Designed and implemented automation and process-control improvements, including an HVAC control modification that reduced electrical power consumption by approximately 10% and a CO_2 pressure-control solution that contributed to increasing plant CO_2 yield from approximately 68% to 72%.

- Served as site administrator and process owner for PepsiCo Manufacturing & Warehousing (M&W) performance management, monitoring Overall Equipment Effectiveness (OEE), beverage-power consumption, CO_2 yield, process losses, and equipment downtime while maintaining beverage-power consumption below the 0.33 kWh per 8-oz-case target.
- Used OEE and downtime analysis to identify recurring equipment failures and production bottlenecks, improving reliability through structured preventive maintenance, rapid breakdown response, rigorous follow-up, cross-functional corrective actions, and equipment-specific training for technicians and operators.

Team Member – Electrical 2017 – 2018

BSRM Wires & Epoxy Ltd.

- Worked in a technically focused electrical and automation role within a large-scale steel manufacturing environment, developing hands-on expertise in PLC-based control, industrial drives, instrumentation, process automation, commissioning, and troubleshooting.
- Performed troubleshooting and fault diagnostics of Siemens STEP 7 PLC systems using SIMATIC Manager and TIA Portal for industrial production and process-control equipment.
- Supported electrical maintenance and process troubleshooting of the Fusion Bonded Epoxy (FBE) coating plant, including shot blasting, induction heating, coating, dust collection, cooling, and water-recirculation systems.
- Installed, commissioned, parameterized, and troubleshot ABB DCS880, ACS550, ACS880, and ACS355 drive systems and associated motors and industrial equipment.
- Installed, calibrated, and maintained industrial instrumentation including RTDs, thermocouples, differential-pressure transmitters, load cells, flow, level, and pressure transmitters, and control valves.
- Designed and implemented an automated rod-counting and rejection system using a Siemens S7-1200 PLC and KTP HMI, and contributed to production-line modifications that increased plant capacity.
- Supported implementation of the Oracle EBS Enterprise Asset Management (EAM) module for preventive maintenance, equipment records, and plant asset management.

Assistant Engineer / Assistant Manager → Deputy Manager – Electrical 2012 – 2017

Jeans 2000 Ltd. (Pacific Jeans Group)

- Progressed to Deputy Manager – Electrical with responsibility for supervising electrical maintenance, automation, utility systems, engineering projects, safety compliance, and technical execution across manufacturing operations.
- Planned, coordinated, and monitored electrical and automation projects, controlling project schedules, budgets, work quality, execution plans, and performance while reporting project status and coordinating contractors and technical teams.
- Served as Project Engineer for installation and acceptance testing of a 4 MVA, 11/0.44 kV substation and for installation of fire-hydrant and fire-detection systems in accordance with NFPA 20, NFPA 70/NEC, and NFPA 72 requirements.
- Supervised installation, commissioning, modification, and maintenance of automated process systems, including Siemens S7-300 PLC-based ETP and reverse-osmosis plants and B&R X20 PLC-based Tonello industrial washing machines.
- Coordinated preventive, periodic, and breakdown maintenance activities in accordance with occupational health and safety requirements, supporting reliable operation of production and utility equipment.
- Performed electrical and fire-safety audits and risk assessments with Bureau Veritas, Accord, and Alliance, coordinating corrective actions to support compliance of plant electrical installations.

Selected Industrial Automation Projects

High-Speed Beverage Production Lines – Transcom Beverages Ltd.
Supported automation, commissioning, troubleshooting, and process optimization of KRONES CONTIFLOW 30/2, CONTIFORM 316 PRO, MODULFILL VFS, Rapid BevTech Carbo-Mix, and KHS INNO-FILL beverage production systems.

Power Distribution, Substation and Energy Management Project – Transcom Beverages Ltd.
Led installation and commissioning of industrial power-distribution infrastructure, including a 4 MVA substation, transformer integration, protection coordination, and associated electrical systems, and implemented an EMS-SCADA system using Modbus TCP/IP to integrate energy meters, PLCs, and plant-level monitoring for real-time energy monitoring, load analysis, and energy-efficiency improvement.

Professional Training & Certifications

KRONES 600 BPM Production Line Training – KRONES Service Team, Germany
ABB DCS880 Operation, Parameterization, Troubleshooting & Maintenance – ABB, India
SIMATIC S7-300/400 Basic PLC Training – Siemens/BSRM
Hot-Fill Quality & Maintenance Training – PepsiCo, India
Sensor Manufacturing & Process Control – University of Colorado Boulder
Safety & Compliance: Electrical Safety, Fire Safety, Risk Assessment, Preventive Maintenance, Bureau Veritas, Accord, Alliance

Education

University of Windsor Windsor, Canada
PhD in Electrical Engineering *Expected Dec 2026*
Chittagong University of Engineering & Technology Bangladesh
BSc in Electrical & Electronics Engineering